Oriented Bounding Box Object Detection

Demonstrate **Ultralytics** using Python: Oriented Bounding Boxes Object Detection effects on images, videos, and real-time detection.

1. Model Introduction

Oriented bounding box object detection, also known as oriented object detection, goes a step further than standard object detection by introducing an additional angle to more accurately locate objects in images.

The output of an oriented object detector is a set of rotated bounding boxes that precisely surround objects in the image, along with class labels and confidence scores for each box. Oriented bounding boxes are particularly useful when objects appear at different angles, such as in aerial images, where traditional axis-aligned bounding boxes might contain unnecessary background.

Simply put, oriented object detection can use **tilted boxes** to precisely surround **tilted objects**, thereby reducing unnecessary background areas and improving detection accuracy.

2. Oriented Bounding Box Object Detection: Image

Use yolo26n-obb_ncnn_model to predict images in the ultralytics26 folder (not images that come with ultralytics).

Enter the code folder:

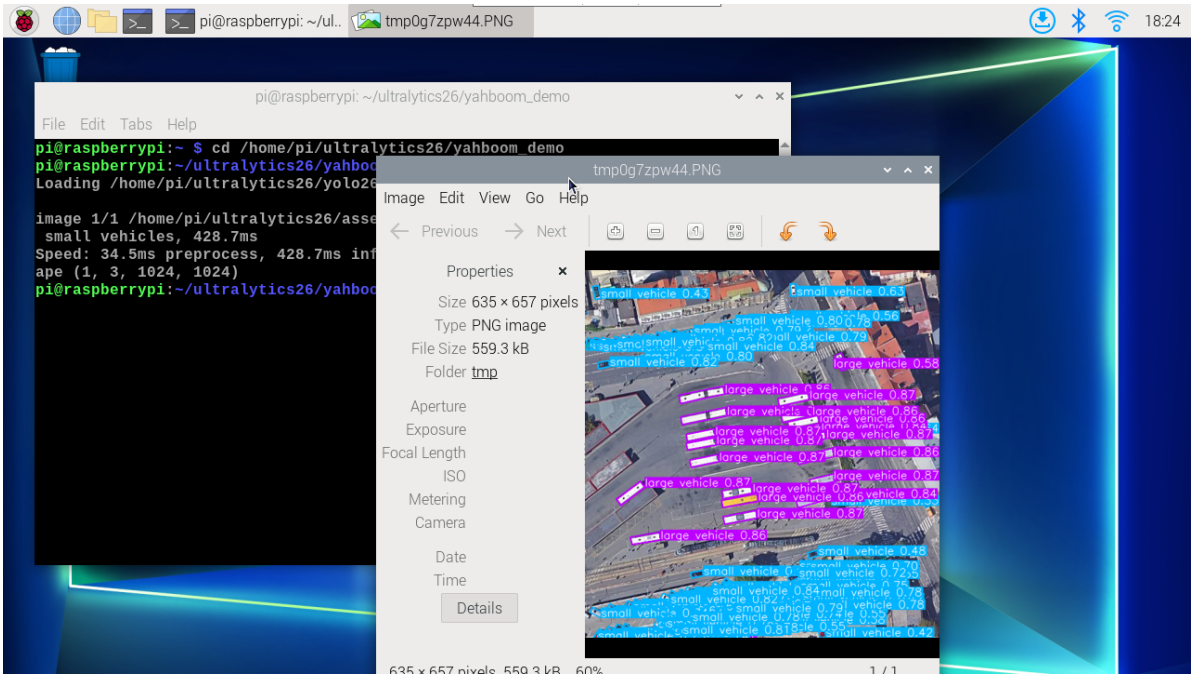
```
cd /home/pi/ultralytics26/yahboom_demo
```

Run the code:

```
python3 05.obb_image.py
```

Effect Preview

yolo recognition output image location: /home/pi/ultralytics26/output/



Example code:

```
from ultralytics import YOLO

# Load a model
model = YOLO("/home/pi/ultralytics26/yolo26n-obb_ncnn_model")
# model = YOLO("/home/pi/ultralytics26/yolo26n-obb.pt")

# Run batched inference on a list of images
results = model("/home/pi/ultralytics26/assets/car.jpg")  # return a list of
Results objects

# Process results list
for result in results:
    boxes = result.boxes  # Boxes object for bounding box outputs
    # masks = result.masks  # Masks object for segmentation masks outputs
    # keypoints = result.keypoints  # Keypoints object for pose outputs
    # probs = result.probs  # Probs object for classification outputs
    obb = result.obb  # Oriented boxes object for OBB outputs
    result.show()  # display to screen
    result.save(filename="/home/pi/ultralytics26/output/car_output.jpg")  # save
to disk
```

3. Oriented Bounding Box Object Detection: Video

Use yolo26n-obb_ncnn_model to predict videos in the ultralytics26 folder (not videos that come with ultralytics).

Enter the code folder:

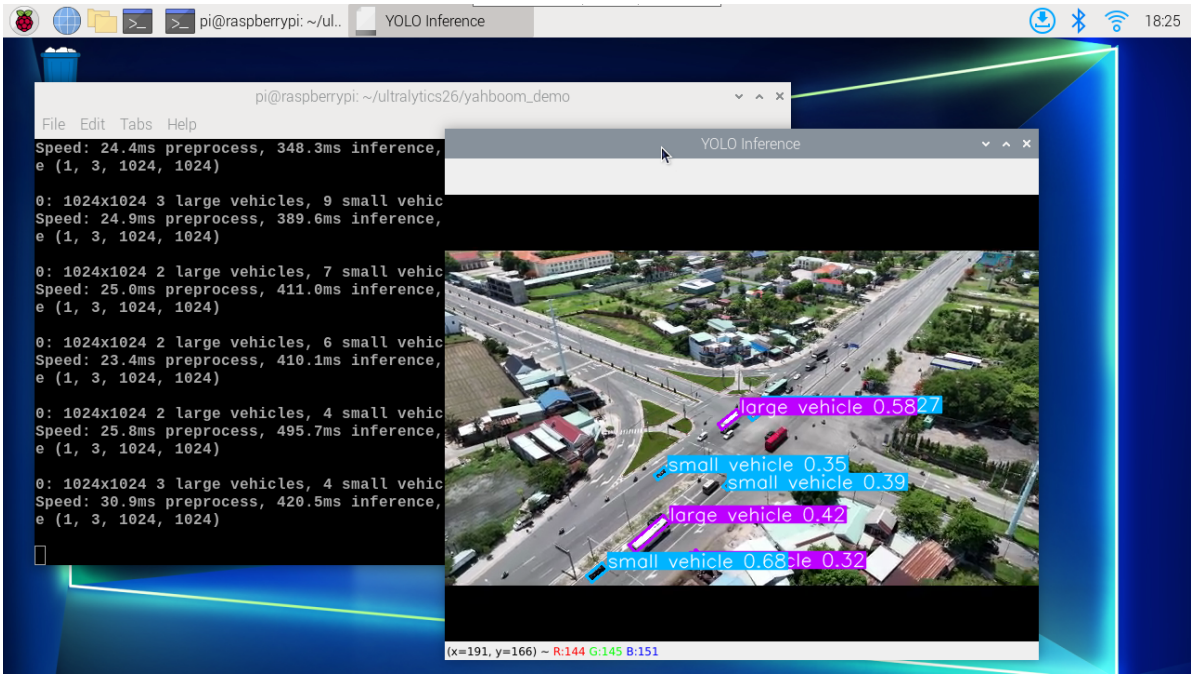
```
cd /home/pi/ultralytics26/yahboom_demo
```

Run the code:

```
python3 05.obb_video.py
```

Effect Preview

yolo recognition output video location: /home/pi/ultralytics26/output/



Example code:

```
import cv2
from ultralytics import YOLO

# Load the YOLO model
model = YOLO("/home/pi/ultralytics26/yolo26n-obb_ncnn_model")
# model = YOLO("/home/pi/ultralytics26/yolo26n-obb.pt")

# Open the video file
video_path = "/home/pi/ultralytics26/videos/street.mp4"
cap = cv2.VideoCapture(video_path)

# Get the video frame size and frame rate
frame_width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
frame_height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = int(cap.get(cv2.CAP_PROP_FPS))

# Define the codec and create a VideoWriter object to output the processed video
output_path = "/home/pi/ultralytics26/output/05.street_output.mp4"
fourcc = cv2.VideoWriter_fourcc(*'mp4v')  # You can use 'XVID' or 'mp4v'
depending on your platform
out = cv2.VideoWriter(output_path, fourcc, fps, (frame_width, frame_height))

# Loop through the video frames
while cap.isOpened():
    # Read a frame from the video
    success, frame = cap.read()

    if success:
```

```
        # Run YOLO inference on the frame
        results = model(frame)

        # Visualize the results on the frame
        annotated_frame = results[0].plot()

        # Write the annotated frame to the output video file
        out.write(annotated_frame)

        # Display the annotated frame
        cv2.imshow("YOLO Inference", cv2.resize(annotated_frame, (640, 480)))

        # Break the loop if 'q' is pressed
        if cv2.waitKey(1) & 0xFF == ord("q"):
            break
    else:
        # Break the loop if the end of the video is reached
        break
# Release the video capture and writer objects, and close the display window
cap.release()
out.release()
cv2.destroyAllWindows()
```

4. Oriented Bounding Box Object Detection: Real-time Detection

4.1 USB Camera

Use yolo26n-obb_ncnn_model to predict USB camera footage.

Enter the code folder:

```
cd /home/pi/ultralytics26/yahboom_demo
```

Run the code:

```
python3 05.obb_camera_usb.py
```

Effect Preview

yolo recognition output video location: /home/pi/ultralytics26/output/

Example code:

```python
import cv2
from ultralytics import YOLO

# Load the YOLO model
model = YOLO("/home/pi/ultralytics26/yolo26n-obb_ncnn_model")
# model = YOLO("/home/pi/ultralytics26/yolo26n-obb.pt")

# Open the camera
cap = cv2.VideoCapture(0)
cap.set(6, cv2.VideoWriter.fourcc('M', 'J', 'P', 'G'))
cap.set(cv2.CAP_PROP_FRAME_WIDTH, 640)
cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 480)

# Get the video frame size and frame rate
frame_width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
frame_height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = int(cap.get(cv2.CAP_PROP_FPS))

# Define the codec and create a VideoWriter object to output the processed video
output_path = "/home/pi/ultralytics26/output/05.obb_camera_usb.mp4"
fourcc = cv2.VideoWriter_fourcc(*'mp4v')  # You can use 'XVID' or 'mp4v'
depending on your platform
out = cv2.VideoWriter(output_path, fourcc, fps, (frame_width, frame_height))

# Loop through the video frames
while cap.isOpened():
    # Read a frame from the video
    success, frame = cap.read()

    if success:
        # Run YOLO inference on the frame
        results = model(frame)

        # Visualize the results on the frame
        annotated_frame = results[0].plot()
```

```
        # Write the annotated frame to the output video file
        out.write(annotated_frame)

        # Display the annotated frame
        cv2.imshow("YOLO Inference", cv2.resize(annotated_frame, (640, 480)))

        # Break the loop if 'q' is pressed
        if cv2.waitKey(1) & 0xFF == ord("q"):
            break
    else:
        # Break the loop if the end of the video is reached
        break
# Release the video capture and writer objects, and close the display window
cap.release()
out.release()
cv2.destroyAllWindows()
```

4.2 CSI Camera

Use yolo26n-obb_ncnn_model to predict CSI camera footage.

Enter the code folder:

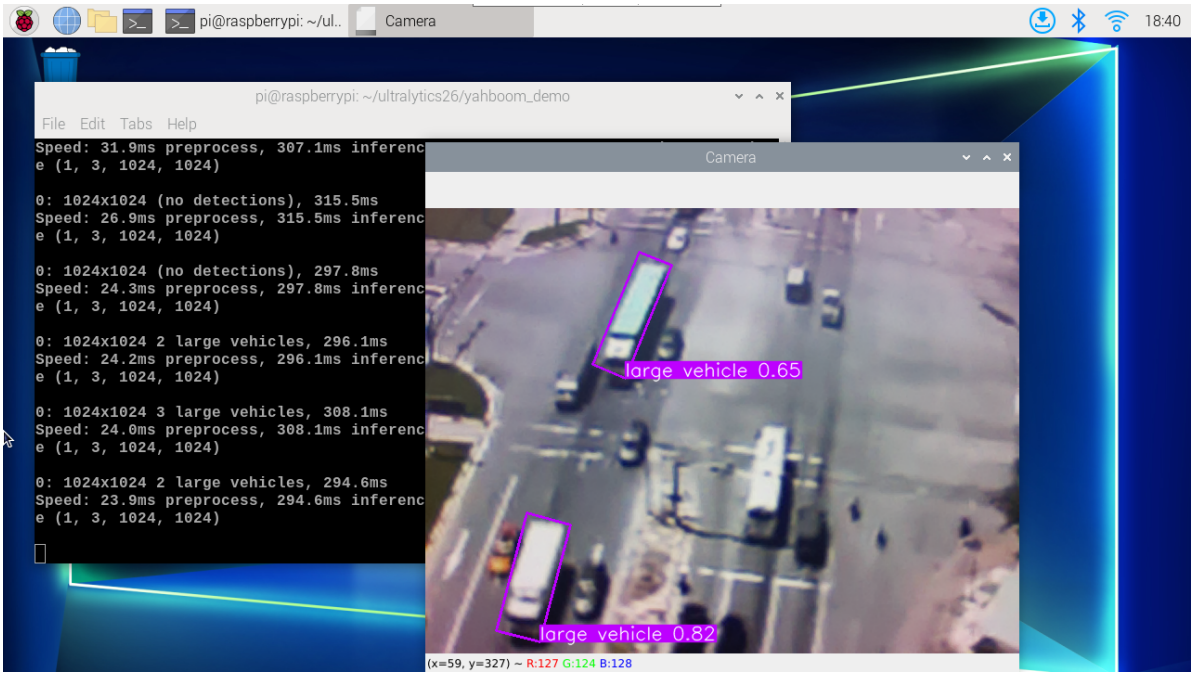
```
cd /home/pi/ultralytics26/yahboom_demo
```

Run the code:

```
python3 05.obb_camera_csi.py
```

Effect Preview

yolo recognition output video location: /home/pi/ultralytics26/output/



Example code:

```
import cv2
```

```python
from picamera2 import Picamera2

from ultralytics import YOLO

# Initialize the Picamera2
picam2 = Picamera2()
picam2.preview_configuration.main.size = (640, 480)
picam2.preview_configuration.main.format = "RGB888"
picam2.preview_configuration.align()
picam2.configure("preview")
picam2.start()

# Load the YOLO26 model
model = YOLO("/home/pi/ultralytics26/yolo26n-obb_ncnn_model")
# model = YOLO("/home/pi/ultralytics26/yolo26n-obb.pt")

# Set up video output
output_path = "/home/pi/ultralytics26/output/05.obb_camera_csi.mp4"
fourcc = cv2.VideoWriter_fourcc(*'mp4v')
out = cv2.VideoWriter(output_path, fourcc, 30, (640, 480))

# Loop through the video frames
while True:
    # Capture frame-by-frame
    frame = picam2.capture_array()

    # Run YOLO inference on the frame
    results = model(frame)

    # Visualize the results on the frame
    annotated_frame = results[0].plot()

    # Write the frame to the video file
    out.write(annotated_frame)

    # Display the resulting frame
    cv2.imshow("Camera", annotated_frame)

    # Break the loop if 'q' is pressed
    if cv2.waitKey(1) == ord("q"):
        break

# Release resources and close windows
picam2.close()
out.release()
cv2.destroyAllWindows()
```

References

[Oriented Bounding Box Object Detection - Ultralytics YOLO Documentation]